MIL-S-46373A(GL)
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SUPERSEDING
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MILITARY SPECIFICATION

STOP WATCH, PARACHUTIST'S

1. SCOPE

1.1 This specification covers a noncontinuous movement, single action timer stop watch, adjusted for operation in one position.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-360 - Time Measuring Instruments: Packaging of

MILITARY

MIL-L-25142 - Luminescent Material, Fluorescent
MIL-B-27497 - Bearing Jewel, Sapphire or Ruby, Synthetic

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6645

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3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced a sample of the stop watch shall be made ready for or submitted to the contracting officer or his authorized representative for examination and tests specified in paragraph 4.2 to determine compliance with the requirements of this specification. The approval of the preproduction sample authorizes the commencement of production but does not relieve the contractor of the responsibility for compliance with all applicable provisions of this specification.

3.2 Design.- The stop watch, adjusted for operation in the horizontal dial up position, shall be designed so that successive depression of the crown will cause the hand to start, stop, and return to zero.

3.2.1 Case.- The case shall be bingeless and shall be of a permanent, lusterless, black, color. The size of the case shall be as specified in Figure 1.

3.2.2 Dial.- The dial shall be secured to the movement from the front by a minimum of two screws. Dial markings, indicator hand design and finish requirements shall be as specified in Table I, and as shown or specified on Figure I.

TABLE I - Dial Marking (Nominal Dimensions)

Markings	Height or Length	Width	Line Thickness
One second graduations	1/16	-	.010
Five second graduations	1/8	-	.031
Triangles	5/32	5/32	-
Numerals	.109	.078	.025
Annular rings	Rings shall be 1/16 apart		.015

3.2.3 Hand.- The indicator hand shall be as specified in Figure 1.

3.2.4 Stem.- The stem shall conform to the dimensional requirements specified in Figure 1.

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3.2.5 Crown.- The crown shall be knurled and shall conform to the dimensional requirements specified in Figure 1.

3.2.6 Movement.- The movement shall be stem wound, non-continuous type, be of jeweled lever type, have not less than seven jewels placed according to good horological practice, and shall be of a size which is in proportion to the size of the case.

3.3 Materials.-

3.3.1 Case.- The case shall be of corrosion-resistant metal.

3.3.2 Crystal.- The crystal shall be of a colorless, transparent, unbreakable, rigid, plastic, material.

3.3.3 Luminescent material.- The fluorescent luminescent material shall conform to type I or III of MIL-L-25142.

3.3.4 Lubricant.- The lubricant used in the watch movement shall permit correct operation of the stop watch within the temperature range of -54°C to +46°C.

3.3.5 Jewels.- The jewels shall conform to the requirements of MIL-B-27497, composition optional.

3.3.6 Mainspring.- The mainspring shall be of an unbreakable alloy and shall be capable of driving the complete movement a minimum of 3 hours without rewinding.

3.3.7 Hairspring.- The hairspring shall be of temperature compensating materials which are not affected in the presence of the magnetic field specified in 3.4.

3.4 Construction.- The stop watch shall conform to the dimensions and details of Figure 1.

3.4.1 Temperature variation.- The stop watch shall be lubricated and conditioned for operation within the temperature range of -54° to +46°C.

3.4.2 Operating force.- The force required to start, stop, and return the indicator hand to zero shall not exceed four pounds.

3.4.3 Winding torque.- The torque to fully wind the mainspring shall not exceed 6 inch ounces. The crown of a fully wound watch shall not move when a torque of 32 inch ounces is applied to the crown.

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3.4.4 Beat.- Unless otherwise specified, (see 6.2) the stop watch shall be a 5-beat watch. For other than a 5-beat watch, the beat designation shall be indicated on the dial or back of case.

3.4.5 Start, stop, and return mechanism.- The mechanism which causes the indicator hand to start, stop, and return to zero shall be capable of withstanding 7500 crown operations (2500 complete cycles of start, stop, and return) when tested in accordance with 4.4.4.

3.4.6 Air leakage.- The amount of air leakage through openings in the case of the stop watch shall not exceed 400 cubic centimeters of air per minute when the stop watch is tested in accordance with 4.4.5.

3.4.7 Shockproof.- The stop watch shall withstand two successive drops from a height of 3 feet to a hardwood floor without any damage to the watch movement other than that which can be corrected by regulating the movement when tested as specified in 4.4.6. A suitable mechanism may be used to simulate shock load.

3.4.8 Magnetic effect.- The stop watch, after exposure to a magnetic field of 60 gaussses for 5 seconds, shall not show a change in accuracy in excess of 4 seconds in 5 minutes when tested as specified in 4.4.7.

3.4.9 Accuracy.- The stop watch shall meet the accuracy requirements specified for room temperature, and -54° and +46°C when tested as specified in 4.4.1.

3.5 Identification marking.- The manufacturer's name or symbol and the serial number shall be marked on the inside of the back. The letters US, not less than 1/4 inches in height, shall be anodized on the back of the case.

3.6 Workmanship.- The workmanship of the stop watch shall conform to the quality established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Preproduction sample inspection.- When required, the preproduction sample shall be inspected and tested for all provisions of this specification applicable to end item inspection.

4.3 Sampling for inspection and acceptance.- Except as otherwise specified herein, sampling for inspection and acceptance shall be in accordance with MIL-STD-105.

4.3.1 Component and material inspection.- Components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or the applicable purchase document.

4.3.2 In-process inspection.- Inspection shall be made during fabrication to determine compliance with design and construction requirements of this specification. The Government reserves the right to exclude from consideration for acceptance any end items for which in-process inspection has indicated nonconformance with design and construction requirements.

4.3.3 Inspection of end item.- The inspection lot shall be formed in accordance with MIL-STD-105. The lot shall be expressed in units of stop watches.

4.3.3.1 Examination for visual characteristics.- The sample unit for this inspection shall be one (1) stop watch. Characteristics to be examined and the defects applicable thereto shall be as follows:

Examine	Defect	Classification Major	Classification Minor
Design and construction	Any characteristic not in accordance with specified requirement	X	
	Any component missing	X	
Finish	Finish not as specified		X
	Scratch through finish to base metal		X
	Color not as specified		X
	Dirt or other foreign matter		X
Workmanship	Any component damaged, cracked or malformed	X	
	Any sharp edge, burr or sliver	X	
	Any component loose or not securely retained	X	

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Examine	Defect	Classification Major	Minor
Workmanship (cont'd)	Any numeral, second graduation line, emphasis mark or annular ring missing	X	
	Indicator hand or any emphasis mark not coated with luminescent material	X	
	Beat mark not indicated where applicable		X
	Luminescent material on indicator hand or emphasis mark peeling or flaking	X	
Identification marking	Missing, incorrect, or illegible		X

4.3.3.2 Dimensional inspection.- The sample unit for this examination is one (1) stop watch. Defects shall be classified as follows:

Examine	Defect	Classification Major
Case diameter	Undersize or oversize	X
Crown diameter	Oversize	X
Stem diameter	Undersize	X
Stem length (depressed position)	Undersize	X
Indicator hand	Radius at tip undersize	X
	Width at center post undersize	X

4.3.3.3 End item testing.- The sample unit for this inspection shall be one stop watch. Each sample selected shall be tested for timing accuracy at room temperature, -54°C and +46°C prior to the conduct of the other tests.

4.4 Tests.-

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4.4.1 Timing accuracy.- Each sample selected shall be tested for timing accuracy at room temperature for each of the following criteria. The stop watch is classified as defective if it exceeds any of the allowed tolerances. A master stop watch with tenths of second graduations shall be used in determining timing accuracy.

Criteria	Allowed tolerance (seconds)
Average error for five 30 second runs	.2
Maximum differences in errors noted during five 30 second runs	.4
Average error for three 5 minute runs	.6
Maximum differences in errors noted during three 5 minute runs	.6

After completion of the tests at room temperature, each sample selected shall be tested at -54°C and +46°C. A 50% increase in the allowed tolerances for room temperature shall be permitted. Each sample shall be conditioned for not less than one hour prior to conduct of the test. The tests shall be performed while the watch is exposed to the required temperature. The sample shall be classified as defective if it exceeds any of the allowed tolerances.

4.4.2 Operating force.- Each sample shall be tested for compliance with the operating force specified in 3.4.2.2, using a force gage calibrated in pounds. The sample is classified as defective if the operating force is greater than 4 pounds.

4.4.3 Winding torque.- Each sample shall be tested for compliance with the winding torque specified in 3.4.2.3, using a winding torque calibrated in inch ounces. The sample is classified as defective if the winding torque to fully wind the mainspring exceeds the 6 inch ounces or if applying a winding torque of 32 inch ounces damages the watch such that it cannot be wound or the stop, start, and return mechanism will not function.

4.4.4 Start, stop, and return mechanism.- Each sample shall be tested for conformance with 3.4.5 by either manual or mechanical means. The indicator hand shall advance for a minimum of 20 seconds before each stop operation. The sample is classified as defective if the number of crown operations is less than 7500 (2500 complete cycles).

4.4.5 Air leakage.- Each sample shall be tested for conformance with 3.4.6, using a pressure differential type apparatus which will develop a pressure differential of one inch of water between the inside and the outside of the watch. The sample is classified as defective if the air leakage exceeds 400 cubic centimeters of air per minute.

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4.4.6 Shockproof.- Each sample shall be tested for conformance with 3.4.7 by applying two successive impulses equivalent to dropping the watch from a height of 3 feet to a hardwood floor. The first impulse shall be applied to the back of the watch, the second to the edge. The sample, after after the second impulse, shall be tested for timing accuracy at room temperature. The sample shall be classified as defective if it exceeds any of the allowed tolerances for timing accuracy at room temperature.

4.4.7 Magnetic effect.- Each sample shall be tested for conformance with 3.4.8 by placing the watch horizontally, face up, in a magnetic field of 60 gausses for 50 seconds. Immediately after removal from the magnetic field the watch shall be allowed to run continuously for 5 minutes and the error, if any, noted. The sample is classified as defective if the error for the 5 minute run is in excess of 4 seconds.

4.4.8 Mainspring.- Each sample shall be tested for conformance with 3.3.6 by allowing a watch with the mainspring fully wound to run continuously for not less than 3 hours or until it stops, depending upon which occurs first. The sample is classified as defective if it fails to run continuously for a minimum of 3 hours.

4.4.9 Hairspring.- Each sample which conforms to the requirements of 3.3.6 and 3.4.8 shall be considered as conforming to 3.3.7.

4.5 Examination of preparation for delivery.- Examination shall be made in two phases; first, examination of the interior package; second, an examination of the container fully prepared for delivery. The sample unit for each phase shall be one container fully prepared for delivery. The lot size shall be the number of containers ready for delivery except that containers need not be sealed or closed. Defects shall be scored as specified below:

Examine	Defect
Marking (interior and exterior container)	Omitted, illegible, or incorrect in size, location and method of application.
Material	Damaged, defective, missing, or not as specified.
Workmanship	Container flaps not completely closed. Seams, corners or manufacturer's joint not taped as specified. Strapping or tape loose. Staples loose or missing. Containers bulging or distorted.

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4.6 Inspection levels and acceptable quality levels (AQL's).- Inspection levels and acceptable quality levels (AQL's), expressed in defects per 100 units shall be as follows:

	Inspection level	AQL's
For examination in 4.3.3.1	II	2.5 Major 4.0 Minor
For examination in 4.3.3.2	II	2.5 Major
For all tests in 4.4 except 4.4.4	S-2	2.5
For examination in 4.5	S-4	4.0

Test requirements are applicable to the individual unit, one determination per unit. The results will be reported as pass or fail. For 4.4.4, select one sample from each lot and test for the 7500 crown operations of 3.4.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A or C.- Each stop watch shall be packaged as specified in PPP-T-360.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A, B or C.- Each stop watch packed as specified in 5.1, shall be packaged as specified in PPP-T-360.

5.3 Marking.- Marking shall be in accordance with PPP-T-360.

6. NOTES

6.1 Intended use.- The parachutist's stop watch is used with the high altitude airdrop system by parachutists to determine the length of time of free fall before delayed opening of the parachute.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Federal stock number.
c. Whether preproduction sample is required (see 3.1).
d. If stop watch is other than 5 beat (see 3.4.4).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 The outside margins of this document have not been marked to indicate where changes have been made as document changes have been extensive in all sections.

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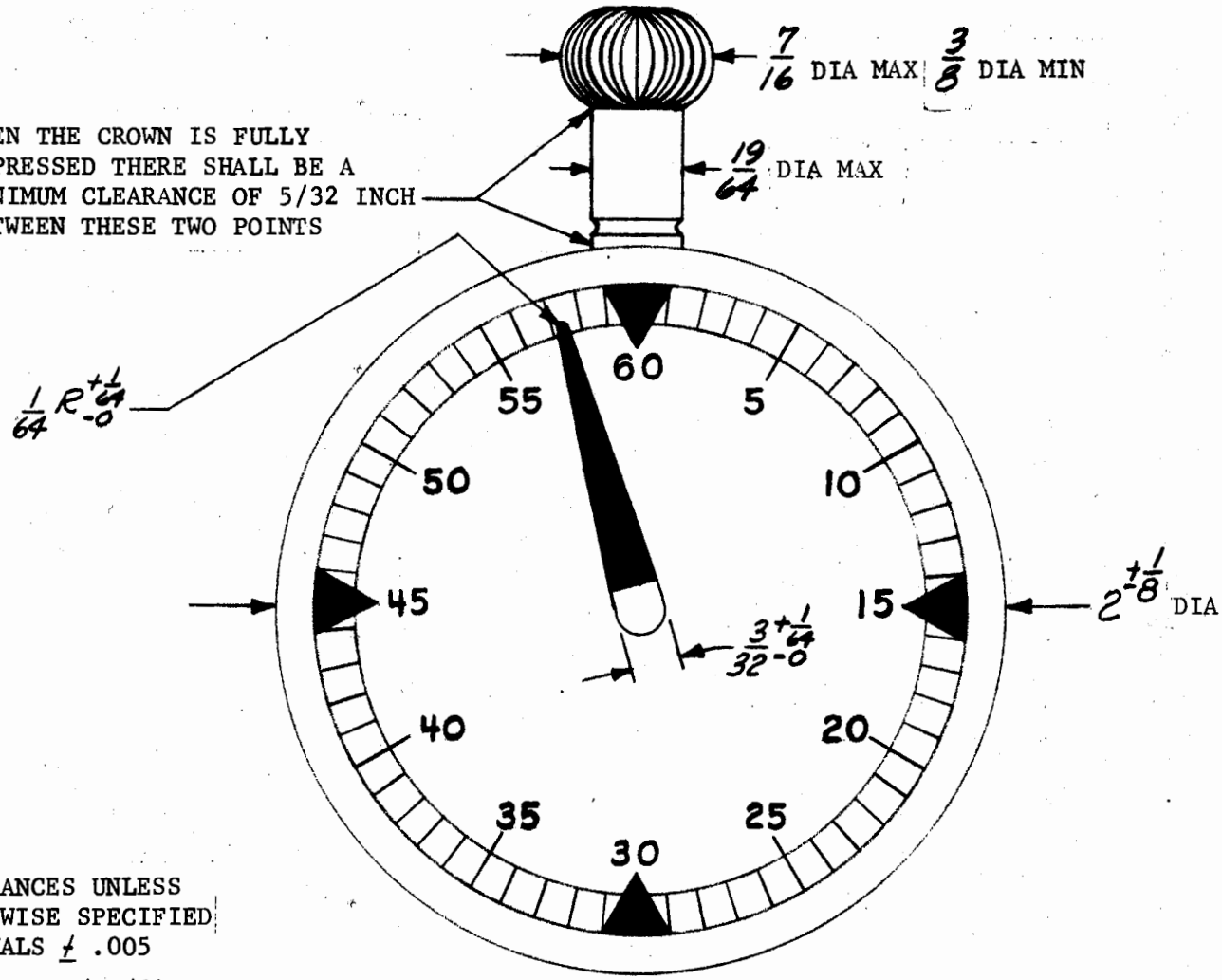
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Project No. 6645-A009

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STOP WATCH, PARACHUTIST'S

FIGURE I

NOTES:

DIAL SHALL BE DULL BLACK.
ANNULAR RINGS, NUMERALS AND SECOND GRADUATION MARKS SHALL BE WHITE.
INDICATOR HAND AND 15 SECOND EMPHASIS MARKS SHALL BE COATED WITH FLUORESCENT LUMINOUS MATERIAL. (SEE PARAGRAPH 3.3.3)
THE INDICATOR HAND SHALL BE COATED FROM THE TIP TO A POINT APPROXIMATELY 1/16 INCH FROM CENTER POST. THAT PORTION OF HAND NOT COATED SHALL BE DULL BLACK.
THE LENGTH OF THE INDICATOR HAND SHALL BE IN PROPORTION TO THE SIZE OF THE DIAL AS SHOWN AND THE TIP SHALL EXTEND OVER THE INSIDE ANNULAR RING 1/64 $\pm \frac{1/64}{0}$

NOT TO SCALE

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